

08C__scaffoldingTest_evaluation

February 19, 2024

```
[1]: #evaluation of RagTag scaffolded genome proteome coverage
#author: Sheng-Kai Hsu
#created: 2024.01.23
#last edited: 2024.02.19
rm(list=ls())
```

```
[44]: library(rtracklayer)
library(Biostrings)
library(limma)
library(scales)
library(LSD)
```

Attaching package: ‘limma’

The following object is masked from ‘package:BiocGenerics’:

plotMA

1 random shortread assemblies

```
[31]: gff_old = readGFF("/workdir/sh2246/p_phyloGWAS/output/scaffoldingTest/miniProt/
↪AN20T005.gff")
gff_new = readGFF("/workdir/sh2246/p_phyloGWAS/output/scaffoldingTest/miniProt/
↪AN20T005.scaffolded.gff")
```

```
[41]: queryFa = readAAStringSet("/workdir/sh2246/p_phyloGWAS/output/Pv-672_v3.
↪0_peptide.fa")
```

```
[42]: queryFa_length=width(queryFa)
names(queryFa_length)=names(queryFa)
```

```
[46]: gff.mrna_old = gff_old[gff_old[,3]=="mRNA",]

queryStat_old = strsplit2(gff.mrna_old$Target," ")
```

```

queryStat_old = as.data.frame(queryStat_old)
colnames(queryStat_old) = c("query", "start", "end")
queryStat_old$range = as.numeric(queryStat_old[,3]) - as.
  ↳ numeric(queryStat_old[,2]) + 1

# query coverage
qcov_old=c()
for (i in unique(queryStat_old[,1])){
  qcov_old=c(qcov_old, queryStat_old[queryStat_old[,1]==i,4]/queryFa_length[i])
}
queryStat_old$coverage = qcov_old

```

```

[47]: gff.mrna_new = gff_new[gff_new[,3]=="mRNA",]

queryStat_new = strsplit2(gff.mrna_new$Target, " ")
queryStat_new = as.data.frame(queryStat_new)
colnames(queryStat_new) = c("query", "start", "end")
queryStat_new$range = as.numeric(queryStat_new[,3]) - as.
  ↳ numeric(queryStat_new[,2]) + 1

# query coverage
qcov_new=c()
for (i in unique(queryStat_new[,1])){
  qcov_new=c(qcov_new, queryStat_new[queryStat_new[,1]==i,4]/queryFa_length[i])
}
queryStat_new$coverage = qcov_new

```

```

[48]: head(queryStat_old)
head(queryStat_new)

```

		query <chr>	start <chr>	end <chr>	range <dbl>	coverage <dbl>
A data.frame: 6 × 5	1	Pavag01G000100.1	1	82	82	0.3727273
	2	Pavag01G000200.1	214	327	114	0.2550336
	3	Pavag01G000200.2	214	327	114	0.2740385
	4	Pavag01G000500.3	124	501	378	0.2988142
	5	Pavag01G000500.1	323	700	378	0.2573179
	6	Pavag01G000500.2	323	700	378	0.2581967
		query <chr>	start <chr>	end <chr>	range <dbl>	coverage <dbl>
A data.frame: 6 × 5	1	Pavag01G000100.1	1	82	82	0.3727273
	2	Pavag01G000200.1	134	442	309	0.6912752
	3	Pavag01G000200.2	134	413	280	0.6730769
	4	Pavag01G000500.3	1	1265	1265	1.0000000
	5	Pavag01G000500.1	33	1469	1437	0.9782165
	6	Pavag01G000500.2	33	1464	1432	0.9781421

```
[54]: queryStat_old_primary = queryStat_old[gff.mrna_old$Rank == 1,]  
      queryStat_new_primary = queryStat_new[gff.mrna_new$Rank == 1,]
```

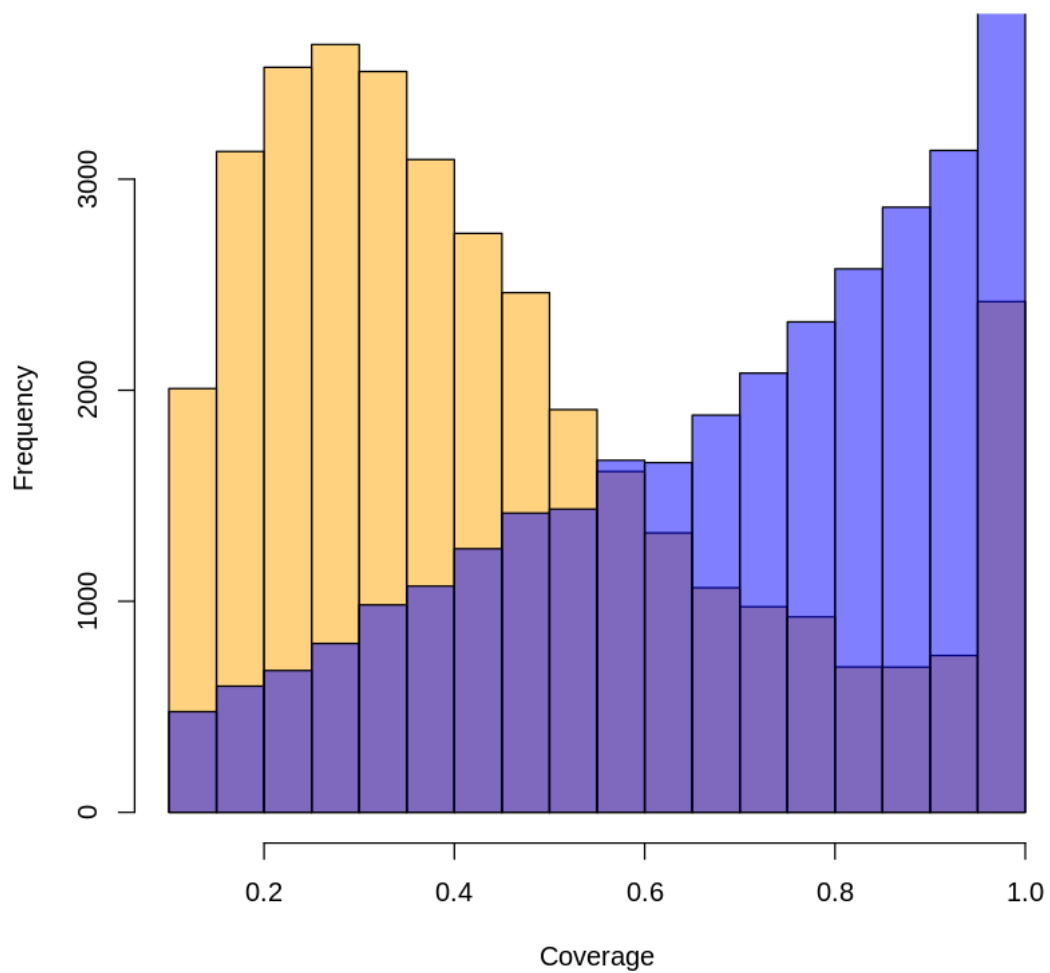
```
[56]: rownames(queryStat_old_primary) = queryStat_old_primary[,1]  
      rownames(queryStat_new_primary) = queryStat_new_primary[,1]
```

```
[59]: dim(queryStat_old_primary)  
      dim(queryStat_new_primary)
```

```
1. 36466 2. 5
```

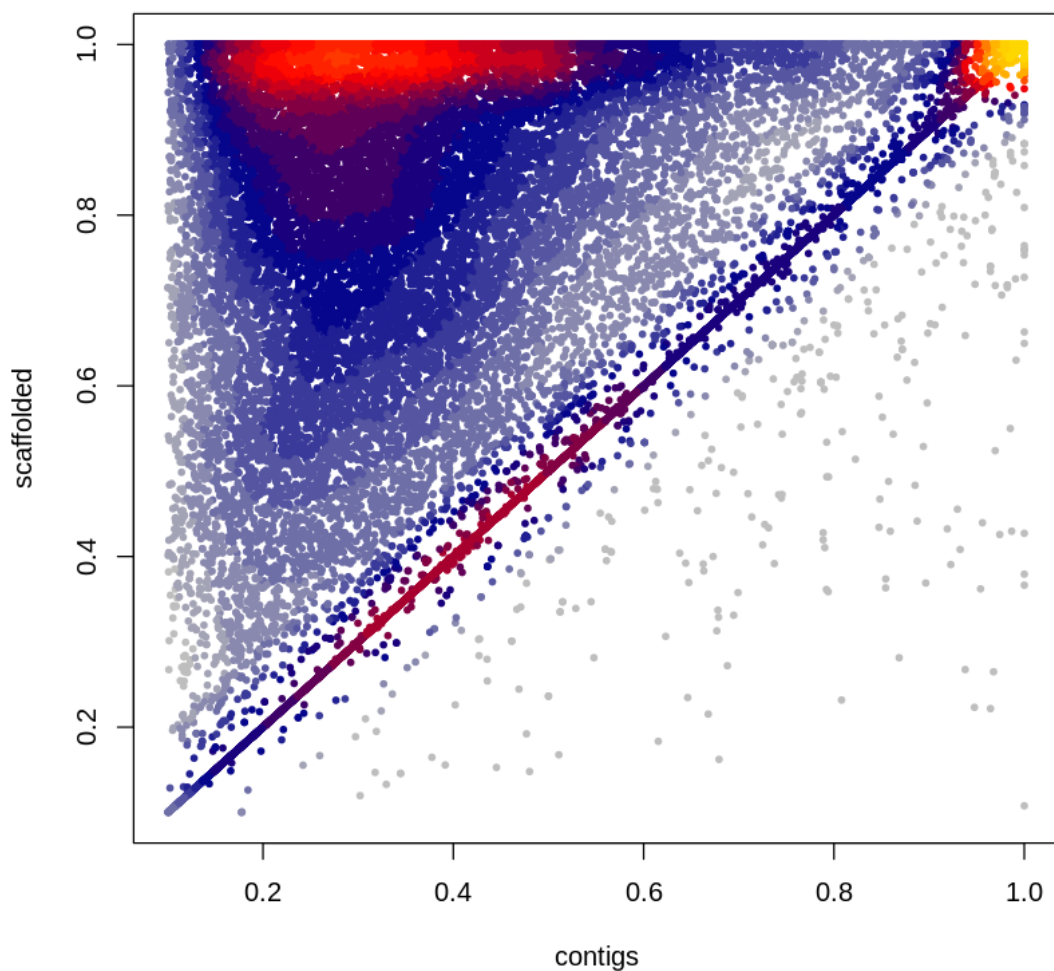
```
1. 36997 2. 5
```

```
[171]: options(repr.plot.width=7, repr.plot.height=7)  
hist(queryStat_old_primary$coverage,xlab = "Coverage",main = "", col = "orange",  
      alpha(.5))  
hist(queryStat_new_primary$coverage,add = T, col = "blue",.5))
```



```
[60]: common_anchor = ⌋
↪ intersect(rownames(queryStat_old_primary), rownames(queryStat_new_primary))
```

```
[88]: heatmap(queryStat_old_primary[common_anchor,]$coverage, queryStat_new_primary[common_anchor,]$coverage,
           xlab = "contigs", ylab = "scaffolded", main = "")
```



```
[102]: badID = queryStat_new_primary[common_anchor,][which(queryStat_new_primary[common_anchor,]$coverage-
      ↪-0.5),1]
length(badID)
gff.mrna_old[strsplit2(gff.mrna_old$Target," ")[,1]%in%badID,]
gff.mrna_new[strsplit2(gff.mrna_new$Target," ")[,1]%in%badID&gff.mrna_new$Rank_
      ↪==1,]
```

18

DataFrame with 18 rows and 18 columns

	seqid	source	type	start	end	score	strand
	<factor>	<factor>	<factor>	<integer>	<integer>	<numeric>	<character>
1	k141_182507	miniprot	mRNA	155	1609	798	-

2	k141_471503	miniprot	mRNA	1	363	58	-
3	k141_328505	miniprot	mRNA	206	1012	462	-
4	k141_72599	miniprot	mRNA	389	1852	692	+
5	k141_723445	miniprot	mRNA	14	1714	740	-
...	...	...	...	...	...	...	...
14	k141_276017	miniprot	mRNA	1566	2189	473	+
15	k141_449893	miniprot	mRNA	166	850	279	-
16	k141_416619	miniprot	mRNA	97	1356	548	-
17	k141_337988	miniprot	mRNA	45	994	372	-
18	k141_116157	miniprot	mRNA	10	788	274	+
	phase	ID	Rank	Identity	Positive		
	<integer>	<character>	<character>	<character>	<character>		
1	NA	MP001778	1	0.3697	0.5383		
2	NA	MP001937	1	0.3488	0.3953		
3	NA	MP001998	1	0.3532	0.5465		
4	NA	MP003286	1	0.5113	0.6656		
5	NA	MP005437	1	0.3030	0.5026		
...	...	...	...	...	...		
14	NA	MP030328	1	0.4605	0.5907		
15	NA	MP033229	1	0.6092	0.7356		
16	NA	MP033398	1	0.3788	0.5127		
17	NA	MP036687	1	0.3507	0.5521		
18	NA	MP037629	1	0.4505	0.5659		
	Target	Parent	Donor	Frameshift	Acceptor		
	<character>	<CharacterList>	<character>	<character>	<character>		
1	Pavag01G163200.1	8 526	NA	NA	NA		
2	Pavag01G179200.1	22 ..	NA	NA	NA		
3	Pavag01G184000.1	8 256	NA	NA	NA		
4	Pavag01G301900.1	1 302	NA	NA	NA		
5	Pavag01G489200.1	132..	NA	NA	NA		
...	...	...	...	...	...		
14	Pavag09G096300.1	19 ..	NA	NA	NA		
15	Pavag10G104300.1	1 87	NA	NA	NA		
16	Pavag10G122500.1	24 ..	NA	2	NA		
17	PavagK184500.1	34 317	NA	NA	NA		
18	PavagK296900.1	6 179	NA	NA	NA		
	StopCodon						
	<character>						
1	NA						
2	NA						
3	NA						
4	NA						
5	NA						
...	...						
14	NA						
15	NA						
16	NA						
17	NA						

18

NA

DataFrame with 18 rows and 18 columns

	seqid	source	type	start	end	score	strand
	<factor>	<factor>	<factor>	<integer>	<integer>	<numeric>	<character>
1	Chr01_RagTag	miniprot	mRNA	5890273	5891756	833	+
2	Chr03_RagTag	miniprot	mRNA	4624130	4624234	55	+
3	Chr01_RagTag	miniprot	mRNA	6844293	6845779	455	+
4	Chr01_RagTag	miniprot	mRNA	12918366	12918761	567	-
5	Chr01_RagTag	miniprot	mRNA	19596986	19597372	607	+
...	...	...	...	...	...	...	...
14	Chr09_RagTag	miniprot	mRNA	5052656	5054288	471	+
15	Chr10_RagTag	miniprot	mRNA	3488924	3489022	164	-
16	Chr10_RagTag	miniprot	mRNA	4528918	4529217	413	-
17	Chr05_RagTag	miniprot	mRNA	15228987	15229262	329	-
18	Chr03_RagTag	miniprot	mRNA	4275445	4277640	237	+
	phase	ID	Rank	Identity	Positive		
	<integer>	<character>	<character>	<character>	<character>		
1	NA	MP001790	1	0.6739	0.6884		
2	NA	MP001950	1	0.4857	0.4857		
3	NA	MP002008	1	0.8073	0.8807		
4	NA	MP003309	1	0.8030	0.9015		
5	NA	MP005455	1	0.8682	0.9457		
...	...	...	...	...	...		
14	NA	MP030329	1	0.9670	0.9890		
15	NA	MP033222	1	0.9394	0.9697		
16	NA	MP033388	1	0.8300	0.9000		
17	NA	MP036739	1	0.7500	0.8261		
18	NA	MP037695	1	0.6092	0.6782		
	Target	Parent	Frameshift	Acceptor	StopCodon		
	<character>	<CharacterList>	<character>	<character>	<character>		
1	Pavag01G163200.1 301..		1	NA	NA		
2	Pavag01G179200.1 1 35			NA	NA		NA
3	Pavag01G184000.1 1 109			NA	NA		NA
4	Pavag01G301900.1 179..			NA	NA		NA
5	Pavag01G489200.1 432..			NA	NA		NA
...	...	...	...	...	...		...
14	Pavag09G096300.1 101..			NA	NA		NA
15	Pavag10G104300.1 7 39			NA	NA		NA
16	Pavag10G122500.1 23 ..			NA	NA		NA
17	PavagK184500.1 1 92			NA	NA		NA
18	PavagK296900.1 100 179			NA	NA		NA
	Donor						
	<character>						
1	NA						
2	NA						
3	NA						
4	NA						

```

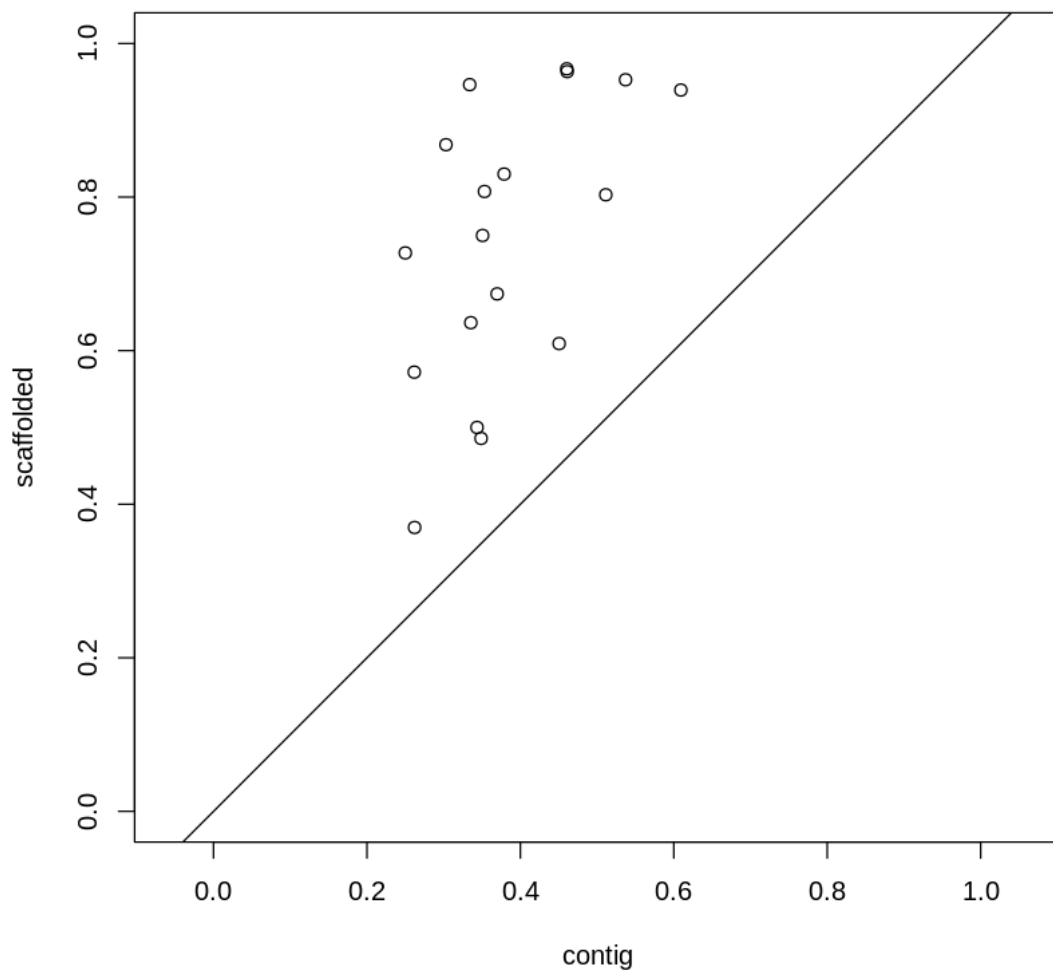
5          NA
...      ..
14        NA
15        NA
16        NA
17        NA
18        NA

```

```

[106]: plot(as.numeric(gff.mrna_old[strsplit2(gff.mrna_old$Target,"␣
↪")][,1]%in%badID,11)),
          as.numeric(gff.mrna_new[strsplit2(gff.mrna_new$Target,"␣
↪")][,1]%in%badID&gff.mrna_new$Rank ==1,11)),
          asp =1,xlim = c(0,1),ylim = c(0,1),xlab = "contig",ylab = "scaffolded")
abline(a =0, b = 1)

```



```
[125]: head(intersect(queryStat_old[duplicated(queryStat_old[,1]),1],
                      ↵
                      ↪common_anchor[which(queryStat_new_primary[common_anchor,]$coverage>0.95&
                                   queryStat_old_primary[common_anchor,]$coverage< 0.5)]))
```

1. 'Pavag01G022300.1' 2. 'Pavag01G022500.2' 3. 'Pavag01G026100.1' 4. 'Pavag01G030100.1'
5. 'Pavag01G058800.1' 6. 'Pavag01G068200.1'

```
[129]: gff_old[grep("Pavag01G022300.1",gff_old$Target),]
       gff_new[grep("Pavag01G022300.1",gff_new$Target),]
```

DataFrame with 5 rows and 18 columns

	seqid	source	type	start	end	score	strand
	<factor>	<factor>	<factor>	<integer>	<integer>	<numeric>	<character>
1	k141_183403	miniprot	mRNA	130	357	219	+
2	k141_183403	miniprot	CDS	130	198	110	+
3	k141_183403	miniprot	CDS	277	354	109	+
4	k141_50253	miniprot	mRNA	560	730	223	-
5	k141_50253	miniprot	CDS	560	730	223	-

	phase	ID	Rank	Identity	Positive
	<integer>	<character>	<character>	<character>	<character>
1	NA	MP000245	1	0.9388	0.9388
2	0	NA	1	1.0000	NA
3	0	NA	1	0.8846	NA
4	NA	MP000246	2	0.8596	0.8596
5	0	NA	2	0.8596	NA

	Target	Parent	Donor	Frameshift	Acceptor
	<character>	<CharacterList>	<character>	<character>	<character>
1	Pavag01G022300.1 75 ..		NA	NA	NA
2	Pavag01G022300.1 75 97	MP000245	NA	NA	NA
3	Pavag01G022300.1 98 ..	MP000245	NA	NA	NA
4	Pavag01G022300.1 1 56		NA	NA	NA
5	Pavag01G022300.1 1 56	MP000246	NA	NA	NA

	StopCodon
	<character>
1	NA
2	NA
3	NA
4	NA
5	NA

DataFrame with 5 rows and 18 columns

	seqid	source	type	start	end	score	strand
	<factor>	<factor>	<factor>	<integer>	<integer>	<numeric>	<character>
1	Chr01_RagTag	miniprot	mRNA	796349	799316	505	-
2	Chr01_RagTag	miniprot	CDS	799146	799316	223	-
3	Chr01_RagTag	miniprot	CDS	797121	797176	68	-

```

4 Chr01_RagTag miniprot CDS 796508 796574 105 -
5 Chr01_RagTag miniprot CDS 796352 796429 109 -
      phase      ID      Rank      Identity      Positive
<integer> <character> <character> <character> <character>
1      NA      MP000240      1      0.8871      0.8871
2      0      NA      1      0.8596      NA
3      0      NA      1      0.8421      NA
4      1      NA      1      1.0000      NA
5      0      NA      1      0.8846      NA
      Target      Parent      Frameshift      Acceptor      StopCodon
<character> <CharacterList> <character> <character> <character>
1 Pavag01G022300.1 1 123      NA      NA      NA
2 Pavag01G022300.1 1 56      MP000240      NA      NA      NA
3 Pavag01G022300.1 57 74      MP000240      NA      NA      NA
4 Pavag01G022300.1 75 97      MP000240      NA      NA      NA
5 Pavag01G022300.1 98 ..      MP000240      NA      NA      NA
      Donor
<character>
1      NA
2      NA
3      NA
4      NA
5      NA

```

```

[89]: # case study
agp = read.table("/workdir/sh2246/p_phyloGWAS/output/scaffoldingTest/AN20T005/
↳ragtag.scaffold.agp",skip = 2)

```

```

[127]: agp[agp$V6%in%c("k141_183403","k141_50253"),]

```

	V1	V2	V3	V4	V5	V6	V7	V8	
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>	
A data.frame: 2 × 9	2199	Chr01_RagTag	795412	796705	2199	W	k141_183403	1	1294
	2205	Chr01_RagTag	798587	799432	2205	W	k141_50253	1	846

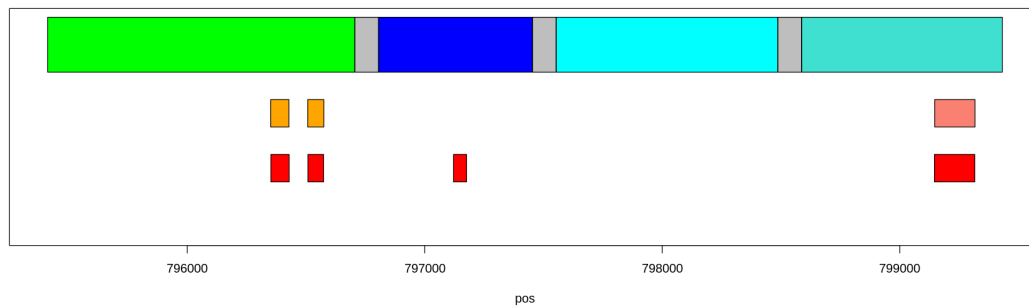
```

[128]: agp[2199:2205,]

```

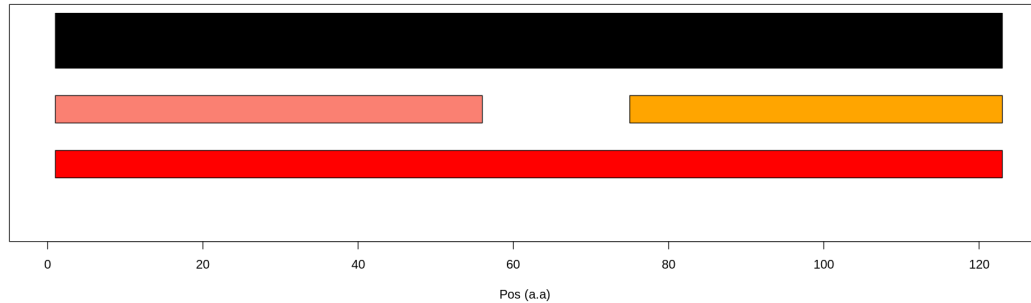
	V1	V2	V3	V4	V5	V6	V7	V8	
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>	
A data.frame: 7 × 9	2199	Chr01_RagTag	795412	796705	2199	W	k141_183403	1	1294
	2200	Chr01_RagTag	796706	796805	2200	U	100	scaffold	yes
	2201	Chr01_RagTag	796806	797453	2201	W	k141_59168	1	648
	2202	Chr01_RagTag	797454	797553	2202	U	100	scaffold	yes
	2203	Chr01_RagTag	797554	798486	2203	W	k141_393521	1	933
	2204	Chr01_RagTag	798487	798586	2204	U	100	scaffold	yes
	2205	Chr01_RagTag	798587	799432	2205	W	k141_50253	1	846

```
[166]: options(repr.plot.width=15, repr.plot.height=5)
plot(unlist(agp[2199:2205,2:3]),rep(1,14),ylim = c(0.6,1),type = 'n',xlab = "pos",ylab = "",yaxt = 'n')
rect(xleft = agp[2199:2205,2],xright = agp[2199:2205,3],ybottom = 0.9,ytop = 1,
     col = c("green","grey","blue","grey","cyan","grey","turquoise"))
rect(xleft = gff_new[grepl("Pavag01G022300.",
↪1",gff_new$Target)&gff_new$type=="CDS",4],
     xright = gff_new[grepl("Pavag01G022300.",
↪1",gff_new$Target)&gff_new$type=="CDS",5],
     ybottom = 0.7,ytop = .75,
     col = "red")
rect(xleft = -gff_old[grepl("Pavag01G022300.",
↪1",gff_old$Target)&gff_old$type=="CDS",4][1:2]+796705,
     xright = -gff_old[grepl("Pavag01G022300.",
↪1",gff_old$Target)&gff_old$type=="CDS",5][1:2]+796705,
     ybottom = 0.8,ytop = .85,
     col = "orange")
rect(xleft = gff_old[grepl("Pavag01G022300.",
↪1",gff_old$Target)&gff_old$type=="CDS",4][3]+798587,
     xright = gff_old[grepl("Pavag01G022300.",
↪1",gff_old$Target)&gff_old$type=="CDS",5][3]+798587,
     ybottom = 0.8,ytop = .85,
     col = "salmon")
```



```
[168]: options(repr.plot.width=15, repr.plot.height=5)
plot(0,0,xlim= c(0,queryFa_length["Pavag01G022300.1"]),ylim= c(0.6,1),type = 'n',xlab = "Pos (a.a)",
     ylab = "", yaxt = 'n')
rect(xleft = 1,xright = queryFa_length["Pavag01G022300.1"],ybottom = 0.9,ytop = 1,
     col = c("black"))
rect(xleft = queryStat_new[queryStat_new[,1]=="Pavag01G022300.1",2],
     xright = queryStat_new[queryStat_new[,1]=="Pavag01G022300.1",3],
     ybottom = 0.7,ytop = .75,col = c("red"))
```

```
rect(xleft = queryStat_old[queryStat_old[,1]=="Pavag01G022300.1",2],
     xright = queryStat_old[queryStat_old[,1]=="Pavag01G022300.1",3],
     ybottom = 0.8,ytop = .85,col = c("orange","salmon"))
```



```
[173]: aln = read.delim("/workdir/sh2246/p_phyloGWAS/output/scaffoldingTest/AN20T005/
      ↪ragtag.scaffold.asm.paf",header = F)
```

```
[174]: head(aln)
```

A data.frame: 6 × 18

	V1 <chr>	V2 <int>	V3 <int>	V4 <int>	V5 <chr>	V6 <chr>	V7 <int>	V8 <int>	V9 <int>
1	k141_395744	211	7	210	-	Chr04	68658214	41506723	415
2	k141_395744	211	7	183	+	Chr09	59416394	4164048	416
3	k141_395744	211	29	210	-	Chr02	77742459	5401457	540
4	k141_395744	211	7	162	+	Chr08	62686529	23821582	238
5	k141_395744	211	29	210	-	Chr08	62686529	12681648	126
6	k141_395744	211	7	183	+	Chr05	71854669	17785560	177

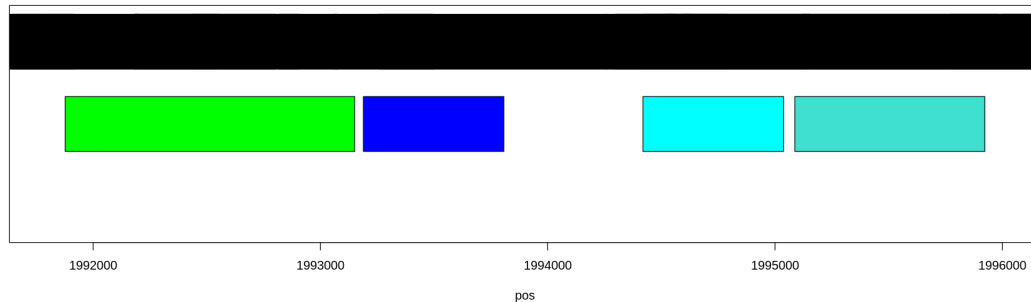
```
[178]: target_aln =
      ↪aln[aln[,1]%in%c("k141_183403","k141_59168","k141_393521","k141_50253"),][1:
      ↪4,]
target_aln
```

A data.frame: 4 × 18

	V1 <chr>	V2 <int>	V3 <int>	V4 <int>	V5 <chr>	V6 <chr>	V7 <int>	V8 <int>
220260	k141_50253	846	14	843	+	Chr01	80884392	1995087
2737254	k141_59168	648	17	635	+	Chr01	80884392	1993189
2880604	k141_183403	1294	10	1284	-	Chr01	80884392	1991877
6397458	k141_393521	933	183	802	-	Chr01	80884392	1994419

```
[186]: TE = read.table("/workdir/sh2246/p_phyloGWAS/data/Sbicolor_313_v3.1.
      ↪repeatmasked_assembly_v3.0.gff3",header = F)
```

```
[197]: options(repr.plot.width=15, repr.plot.height=5)
plot(0,0,xlim = c(1991800,1996000),ylim = c(0.6,1),type = 'n',xlab = "pos",ylab =
  ↪= "",yaxt = 'n')
rect(xleft = target_aln[,8],xright = target_aln[,9],ybottom = 0.75,ytop = .85,
      col = c("green","blue","cyan","turquoise")[c(4,2,1,3)])
rect(xleft = TE[TE[,6]<10,4],xright = TE[TE[,6]<10,5],ybottom = 0.9,ytop =
  ↪1,col = "black")
```



```
[189]: table(TE[,3])
```

```
similarity
519156
```

2 ploidy test

```
[207]: panandPloidy = read.delim("/workdir/sh2246/p_panAndOGASR/data/
  ↪panand_assembly_sizes.txt")
panandPloidy[grep("tdac",panandPloidy[,1]),1] = c("tdacn","tdacs")
```

```
[208]: panandPloidy[panandPloidy[,2]==T&panandPloidy[,5]=="Diploid",1]
panandPloidy[panandPloidy[,2]==F&panandPloidy[,5]=="Diploid",1]
panandPloidy[panandPloidy[,2]==T&panandPloidy[,5]=="Tetraploid",1]
panandPloidy[panandPloidy[,2]==F&panandPloidy[,5]=="Tetraploid",1]
panandPloidy[panandPloidy[,2]==T&panandPloidy[,5]=="Hexaploid",1]
panandPloidy[panandPloidy[,2]==F&panandPloidy[,5]=="Hexaploid",1]
```

1. 'avirgi' 2. 'crefra' 3. 'irugos' 4. 'ppanic' 5. 'rtuber' 6. 'smicro' 7. 'sbicol'

1. 'cserru' 2. 'telega' 3. 'ttrian'

1. 'achine' 2. 'etrips' 3. 'rottb'

1. 'atenui' 2. 'ccitra' 3. 'hconto' 4. 'sscopa' 5. 'snutan' 6. 'vcuspi'

'udigit'

1. 'agerar' 2. 'blagur' 3. 'hcompr'

```
[241]: agp_ploidy = read.table("/workdir/sh2246/p_phyloGWAS/output/scaffoldingTest/
      ↪ ploidy/Ac-Pasquet1232-DRAFT-PanAnd-1.0/ragtag.scaffold.agp", skip = 2)
```

```
[242]: head(agp_ploidy)
```

A data.frame: 6 × 9

	V1 <chr>	V2 <int>	V3 <int>	V4 <int>	V5 <chr>	V6 <chr>	V7 <chr>	V8 <chr>
1	ctg_1_RagTag	1	710	1	W	ctg_1:15282-15992	1	710
2	ctg_1_RagTag	711	810	2	U	100	scaffold	yes
3	ctg_1_RagTag	811	1520	3	W	ctg_1:16213-16923	1	710
4	ctg_1_RagTag	1521	1620	4	U	100	scaffold	yes
5	ctg_1_RagTag	1621	2330	5	W	ctg_1:16794-17504	1	710
6	ctg_1_RagTag	2331	2430	6	U	100	scaffold	yes

```
[243]: sum(agp_ploidy[,5]=="W"&gsub("_RagTag","",agp_ploidy[,1])=="strsplit2(agp_ploidy[,6],":
      ↪")[,1])
```

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```
[244]: contigPos = strsplit2(agp_ploidy[agp_ploidy[,5]=="W",6],":")
```

```
[245]: contigPos = cbind(contigPos[,1],strsplit2(contigPos[,2],"-"))
```

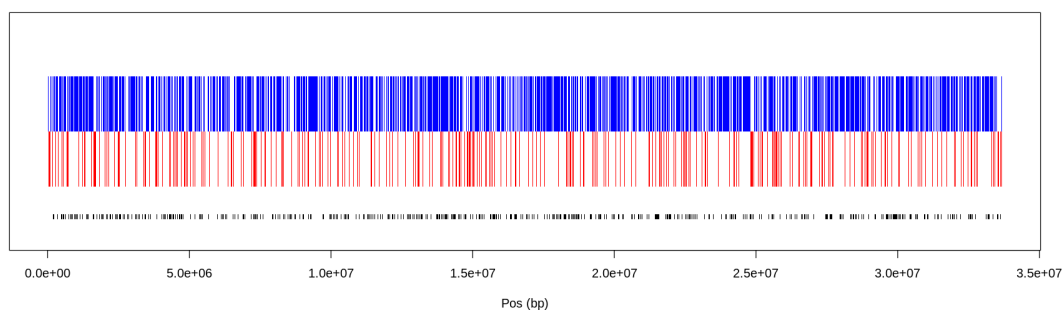
```
[246]: contigPos = cbind(contigPos,grep1("RagTag",agp_ploidy[agp_ploidy[,5]=="W",1]))
```

```
[250]: a = contigPos[contigPos[,1]=="ctg_1",]
```

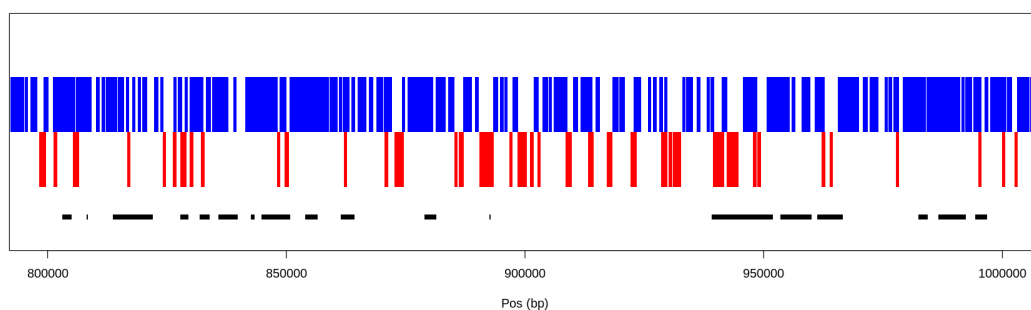
```
[253]: gff_Ac = readGFF("/workdir/sh2246/p_evolBNI/data/annotations_final/
      ↪ 02_Ac-Pasquet1232-DRAFT-PanAnd-1.0_final.gff3")
```

```
[256]: gff_Ac_ctg1 = gff_Ac[gff_Ac[,1]=="ctg_1"&gff_Ac[,3]=="gene",]
```

```
[259]: plot(NA,xlim = c(0, max(as.numeric(a[,3]))),ylim = c(0,1),yaxt = "n",xlab = "Pos_
      ↪ (bp)",ylab = "")
rect(xleft = as.numeric(a[,2]),xright = as.numeric(a[,3]),
     ybottom = ifelse(a[,4],0.5,0.25),ytop = ifelse(a[,4],0.75,0.5),border = NA,
     col = ifelse(a[,4],"blue","red"))
rect(xleft = gff_Ac_ctg1[,4],xright = gff_Ac_ctg1[,5],
     ybottom = 0.1,ytop = .125,border = NA,
     col = "black")
```



```
[262]: plot(NA,xlim = c(8e5, 1e6),ylim =c(0,1),yaxt = "n",xlab = "Pos (bp)",ylab = "")
rect(xleft = as.numeric(a[,2]),xright = as.numeric(a[,3]),
      ybottom = ifelse(a[,4],0.5,0.25),ytop = ifelse(a[,4],0.75,0.5),border = NA,
      col = ifelse(a[,4],"blue","red"))
rect(xleft = gff_Ac_ctg1[,4],xright = gff_Ac_ctg1[,5],
      ybottom = 0.1,ytop = .125,border = NA,
      col = "black")
```



```
[257]: head(gff_Ac_ctg1)
```

DataFrame with 6 rows and 16 columns

	seqid	source	type	start	end	score	strand
	<factor>	<factor>	<factor>	<integer>	<integer>	<numeric>	<character>
1	ctg_1	PanAnd	gene	165372	170719	NA	+
2	ctg_1	PanAnd	gene	195610	199299	NA	+
3	ctg_1	PanAnd	gene	204932	222955	NA	-
4	ctg_1	PanAnd	gene	333288	337363	NA	+
5	ctg_1	PanAnd	gene	338954	343283	NA	+
6	ctg_1	PanAnd	gene	347230	349663	NA	+
	phase		ID	biotype	logic_name		Parent

	<integer>	<character>	<character>	<character>	<CharacterList>
1	NA	Ac00001aa000001	protein_coding	BIND_gene	
2	NA	Ac00001aa000002	protein_coding	BIND_gene	
3	NA	Ac00001aa000003	protein_coding	BIND_gene	
4	NA	Ac00001aa000004	protein_coding	BIND_gene	
5	NA	Ac00001aa000005	protein_coding	BIND_gene	
6	NA	Ac00001aa000006	protein_coding	BIND_gene	
	transcript_id	canonical_transcript	rank	protein_id	
	<character>	<character>	<character>	<character>	
1	NA	NA	NA	NA	
2	NA	NA	NA	NA	
3	NA	NA	NA	NA	
4	NA	NA	NA	NA	
5	NA	NA	NA	NA	
6	NA	NA	NA	NA	

[]: